

Cantilever Beam Bending

Companion to the interactive simulator at <https://aimlab.uk/sims/cantilever-bending.html>

Setup and assumptions

A prismatic cantilever of length L and rectangular cross-section $w \times h$ (width w , height h) is clamped at $x = 0$ and carries a transverse point load F at the free end $x = L$. The material is linear elastic with Young's modulus E . We measure the deflection $u(x)$ positive in the direction of the load and adopt the Euler–Bernoulli assumptions: cross-sections remain plane and perpendicular to the beam axis (shear deformation neglected), and deflections are small ($\delta \lesssim 0.1 L$).

Section property

The second moment of area of the rectangle about its horizontal centroidal axis is

$$I = \int_A y^2 dA = w \int_{-h/2}^{h/2} y^2 dy = \frac{wh^3}{12}. \quad (1)$$

Governing equation

Moment–curvature (from plane sections and Hooke's law) gives $M = EI u''$. Equilibrium of a beam slice gives $dM/dx = V$ and $dV/dx = -q$. Between loads $q = 0$, so

$$EI \frac{d^4 u}{dx^4} = 0. \quad (2)$$

Integrating four times introduces four constants:

$$u''' = C_1, \quad u'' = C_1 x + C_2, \quad u' = \frac{1}{2} C_1 x^2 + C_2 x + C_3, \quad u = \frac{1}{6} C_1 x^3 + \frac{1}{2} C_2 x^2 + C_3 x + C_4. \quad (3)$$

Boundary conditions

The clamped end can neither deflect nor rotate; the free end carries no bending moment, and its shear force equals the applied load:

$$u(0) = 0, \quad u'(0) = 0, \quad EI u''(L) = 0, \quad EI u'''(L) = -F. \quad (4)$$

The first two give $C_4 = C_3 = 0$; the last two give $C_1 = -F/EI$ and $C_2 = FL/EI$.

Solution

$$\boxed{u(x) = \frac{Fx^2(3L-x)}{6EI}} \Rightarrow \delta = u(L) = \frac{FL^3}{3EI}, \quad \theta_L = u'(L) = \frac{FL^2}{2EI}. \quad (5)$$

The bending moment is largest at the wall, $M_{\max} = FL$, so the peak bending stress (at the outer fibres, $y = \pm h/2$) is

$$\sigma_{\max} = \frac{M_{\max}(h/2)}{I} = \frac{6FL}{wh^2}. \quad (6)$$

Deflection grows with L^3 and shrinks with E and h^3 : doubling the height is eight times stiffer, doubling the width only twice.

Worked example (simulator defaults)

$E = 2$ GPa, $L = 500$ mm, $w = 25$ mm, $h = 10$ mm, $F = 4$ N:

$$I = \frac{25 \times 10^3}{12} = 2083 \text{ mm}^4, \quad \delta = \frac{4 \times 0.5^3}{3 \times (2 \times 10^9) \times (2.083 \times 10^{-9})} \approx 40 \text{ mm}, \quad (7)$$

with $\sigma_{\max} = 4.8$ MPa. Set the simulator sliders to these values and compare.